

Assignment 1: Simulating Memory – Devices to Systems

ECE2.414: Advanced Memory Circuits and Systems

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1 PART-A: Ngspice

1.1 Task 1: 6T SRAM Read Stability

The 6T SRAM cell was simulated using the 45 nm PTM BSIM4 model. The initial supply voltage was set to $V_{DD} = 1.1$ V, with equal bitline and complementary bitline capacitances. The read disturbance was measured at $t = 2$ ns.

The original value of $C_{BL} = C_{BLB} = 180$ fF did not produce a sufficiently large variation in the read disturbance. Therefore, the bitline capacitance was corrected to 1.8 pF for the subsequent simulations. With this correction, the measured read disturbance at $V_{DD} = 1.1$ V was

$$\Delta V = 80.32198 \text{ mV}.$$

The specified minimum read-margin threshold was 25 mV. Hence,

$$\Delta V = 80.32198 \text{ mV} > 25 \text{ mV},$$

indicating that the cell satisfies the specified read-stability criterion at $V_{DD} = 1.1$ V.

Table 1: 6T SRAM read-stability result at $V_{DD} = 1.1$ V.

Parameter	Value
V_{DD}	1.1 V
$C_{BL} = C_{BLB}$	1.8 pF
Measurement time	2 ns
Read disturbance, ΔV	80.32 mV
Required threshold	25 mV
Result	Pass

1.2 Task 2: Effect of Access Transistor Sizing

The maximum voltage excursion of the internal storage node was measured during the read operation. For the original access-transistor sizing, the maximum disturbance was

$$q_{\max} = 206.6957 \text{ mV}.$$

The access-transistor width was then changed to $0.24 \mu\text{m}$. With this modification, the maximum disturbance increased to

$$q_{\max} = 273.3965 \text{ mV}.$$

Thus, increasing the access-transistor width increases the read disturbance. A wider access transistor has a larger drive strength, which strengthens the connection between the precharged bitline and the internal storage node during the read operation. Consequently, the internal node storing logic “0” is pulled upward more strongly, resulting in a larger voltage disturbance.



Figure 1: q and q_b voltage for 0.16um width



Figure 2: q and q_b voltage for 0.24um width



Figure 3: q and q_b voltage for $0.60 \mu\text{m}$ width

Table 2: Effect of access-transistor width on read disturbance.

Access transistor width	q_{max}
$0.16 \mu\text{m}$	206.70 mV
$0.24 \mu\text{m}$	273.40 mV
$0.60 \mu\text{m}$	562.77 mV

For a sufficiently large access-transistor width, the cell can approach a metastable condition. In the simulation, increasing the width to $0.60 \mu\text{m}$ resulted in the cell being driven to a metastable state. This demonstrates the trade-off between access strength and read stability in a 6T SRAM cell.

1.3 Task 3: Read Disturbance Versus Supply Voltage

The supply voltage was swept from 1.1 V down to 0.6 V in steps of 50 mV. The resulting read disturbance decreased monotonically with decreasing supply voltage.

Table 3: Read disturbance as a function of supply voltage.

V_{DD}	ΔV
1.10 V	80.322 mV
1.05 V	73.612 mV
1.00 V	66.905 mV
0.95 V	60.205 mV
0.90 V	53.518 mV
0.85 V	46.852 mV
0.80 V	40.222 mV
0.75 V	33.653 mV
0.70 V	27.191 mV
0.65 V	20.922 mV
0.60 V	15.007 mV

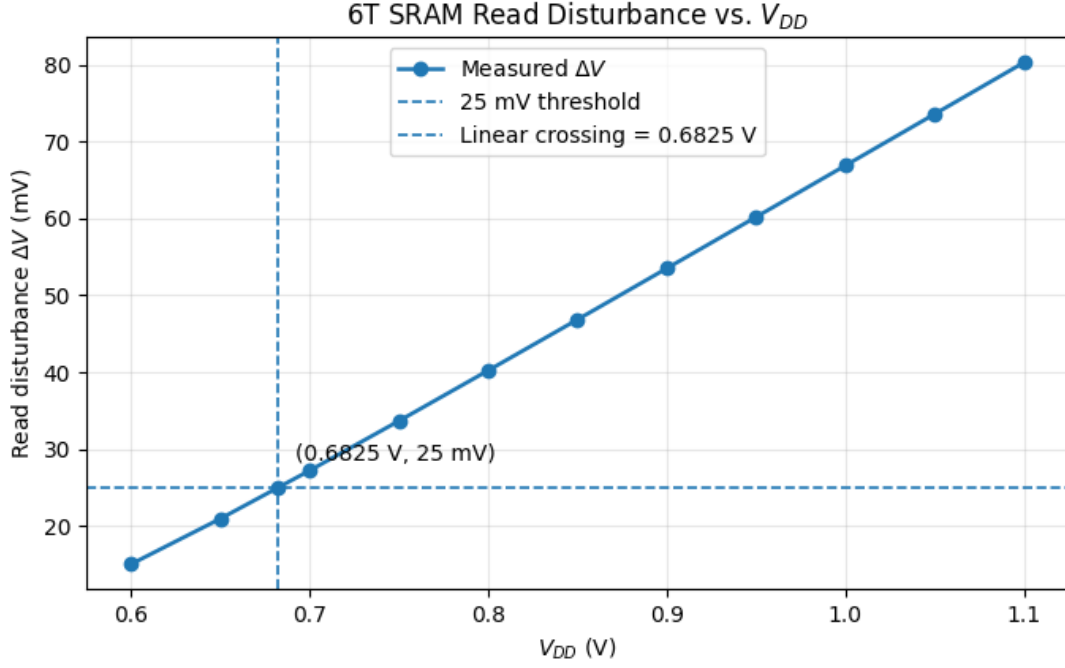


Figure 4: Read disturbance ΔV as a function of supply voltage V_{DD} . The horizontal line represents the 25 mV threshold, and the vertical line indicates the linearly interpolated crossing at $V_{DD} = 0.6825$ V.

The specified threshold is 25 mV. The read disturbance is above this threshold at $V_{DD} = 0.70$ V, but falls below it at $V_{DD} = 0.6825$ V. Therefore, the first supply voltage at which the specified criterion is violated is

$$V_{DD} = 0.6825 \text{ V}.$$

This result shows that reducing the supply voltage progressively reduces the read disturbance. The SRAM therefore has a reduced read-stability margin at low supply voltage.

1.4 Task 4: Temperature Dependence

The SRAM read-stability simulation was repeated at an elevated temperature of 85°C. At this temperature, the measured read disturbance was

$$\Delta V_{85^\circ\text{C}} = 62.69635 \text{ mV}.$$

For comparison, the corresponding value at the nominal temperature was

$$\Delta V_{\text{nominal}} = 80.32198 \text{ mV}.$$

Thus, the read disturbance decreased at the elevated temperature:

$$80.32198 \text{ mV} \longrightarrow 62.69635 \text{ mV}.$$

The measured value at 85°C remains above the specified 25 mV threshold. Hence, the cell still satisfies the given read-disturbance criterion under the simulated high-temperature condition.

Table 4: Temperature dependence of the SRAM read disturbance.

Temperature	ΔV
Nominal	80.322 mV
85°C	62.696 mV

A quantitative comparison with a separately specified sense-amplifier offset was not available from the simulation data. Therefore, no numerical offset margin is claimed here; the measured read disturbance is only compared with the specified 25 mV reference.

2 PART-B: Cacti

2.1 Task 1: 2 MB SRAM Baseline

The 2 MB SRAM cache was evaluated using CACTI 7 with a 64-byte block size, 8-way associativity, one read/write port, four UCA banks, and a 45 nm technology node at 350 K. The optimization target was ED^2 .

The main CACTI results are summarized in Table 5.

Table 5: CACTI 7 results for the 2 MB SRAM cache.

Parameter	Value
Cache capacity	2 MB
Block size	64 B
Associativity	8-way
Number of UCA banks	4
Read/write ports	1
Technology	45 nm
Temperature	350 K
Access time	2.90184 ns
Cycle time	2.65227 ns
Dynamic read energy/access	0.792885 nJ
Dynamic write energy/access	0.851217 nJ
Leakage power	562.632 mW
Cache area	11.476 mm ²

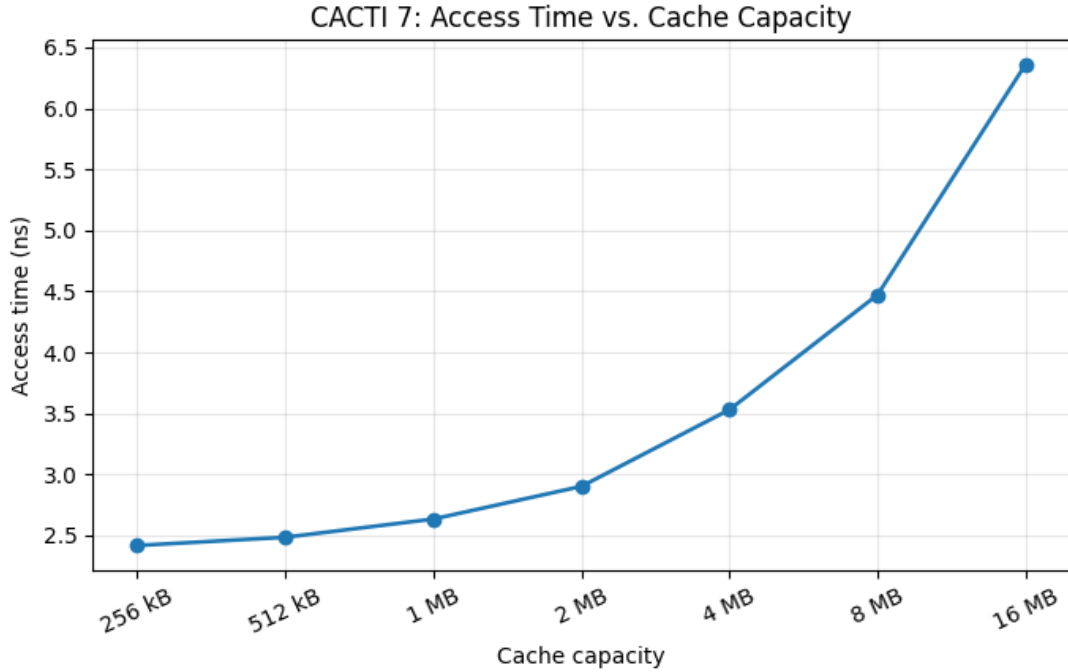


Figure 5: CACTI access time as a function of cache capacity.

The reported access time is 2.90184 ns, while the total dynamic read energy per access is 0.792885 nJ. The total leakage power reported by CACTI is 562.632 mW per bank. For the complete cache, the

data-array leakage is 527.652 mW and the tag-array leakage is 34.9799 mW, which sum to the reported 562.632 mW.

The cache dimensions reported by CACTI are

$$H = 2.21344 \text{ mm}, \quad W = 5.18388 \text{ mm},$$

giving an overall footprint of approximately

$$A = H \times W = 2.21344 \times 5.18388 \approx 11.476 \text{ mm}^2.$$

The data array occupies 10.271 mm^2 , while the tag array occupies 0.387411 mm^2 . The data-array access time is dominated by the H-tree input delay, decoder/wordline delay, bitline delay, and H-tree output delay. In particular, the reported bitline delay is

$$t_{\text{BL}} = 0.406901 \text{ ns}.$$

Thus, the 2 MB SRAM baseline provides an access time of approximately 2.90 ns with a dynamic read energy of 0.793 nJ per access and an area of approximately 11.48 mm^2 .

2.2 Task 2: Capacity Sweep

The cache capacity was swept from 256 kB to 16 MB while keeping the other parameters fixed. As shown in Table 6, access time, read energy, and area all increase with capacity.

Table 6: CACTI capacity sweep.

Capacity	Access (ns)	Read E (pJ)	Area (mm^2)
256 kB	2.4171	658.02	4.965
512 kB	2.4845	678.97	5.946
1 MB	2.6338	716.64	7.600
2 MB	2.9018	792.88	11.474
4 MB	3.5308	930.13	18.458
8 MB	4.4727	1219.75	39.496
16 MB	6.3624	1742.13	74.668

The access-time increase is strongly nonlinear. The main reason is the increasing interconnect delay; at 2 MB, the H-tree alone contributes about 1.36 ns to the access time.

2.3 Task 3: Optimization Objective

The 2 MB cache was evaluated using three optimization objectives. Changing the objective significantly changes the resulting cache organization.

Table 7: Effect of CACTI optimization objective.

Objective	Access (ns)	Read E (pJ)	Area (mm^2)
Pure delay	2.8923	898.45	10.970
Pure area	4.2309	789.99	9.219
ED^2	2.9018	792.88	11.474

The pure-delay configuration is fastest, while pure-area minimizes footprint at the cost of 46% higher access time. The ED^2 objective provides nearly the same delay as pure delay while requiring substantially less read energy.

2.4 Task 4: Hand Check of Bitline Delay

CACTI reports a bitline delay of

$$t_{BL} = 406.9 \text{ ps.}$$

For the 2 MB configuration, the bitline length is $L = 336.38 \mu\text{m}$. Using the distributed RC approximation,

$$t_{RC} \approx 0.38 RCL^2,$$

gives 17.4 ps for the inside-mat wire and 46.6 ps for the local wire, both much smaller than CACTI's value.

The difference occurs because the simple RC calculation considers only the wire. CACTI also accounts for the capacitance of the 512 cells connected to the bitline and the limited current of the SRAM access transistor. Thus, the actual read is current-limited rather than an ideal RC discharge.

3 PART-C: NVsim

3.1 Task 1: STT-MRAM vs. SRAM Baseline

The 2 MB STT-MRAM cache was evaluated using NVSim and compared with the CACTI SRAM baseline from Part B.

Table 8: 2 MB SRAM and STT-MRAM comparison.

Metric	SRAM	STT-MRAM
Hit latency	2.902 ns	3.763 ns
Read energy/access	0.793 nJ	0.734 nJ
Write latency	2.652 ns	10.509 ns
Write energy/access	0.851 nJ	2.063 nJ
Leakage power	2250.5 mW	418.3 mW
Area	11.474 mm ²	0.768 mm ²

Compared with SRAM, STT-MRAM has approximately $1.30\times$ higher read latency but slightly lower read energy. Its major advantages are much lower leakage ($5.4\times$ lower) and area ($14.9\times$ smaller). However, its write latency is about $4\times$ higher and write energy about $2.4\times$ higher.

The STT-MRAM configuration uses a $54F^2$ cell with $R_{on} = 3 \text{ k}\Omega$, $R_{off} = 6 \text{ k}\Omega$, and $200\text{-}\mu\text{A}$ set/reset currents.

3.2 Task 2: STT-MRAM Advantages and Disadvantages

The two major advantages of STT-MRAM are its much lower leakage power and significantly smaller area. Its two major disadvantages are higher write latency and higher write energy.

Table 9: Major STT-MRAM advantages and disadvantages.

Metric	Comparison	Reason
Leakage	$5.4\times$ lower	Non-volatile MTJ cell
Area	$14.9\times$ smaller	$54F^2$ cell
Write latency	$\sim 4\times$ higher	10 ns write pulse
Write energy	$\sim 2.4\times$ higher	$200\text{-}\mu\text{A}$ write current

The low leakage and compact cell are the main benefits of STT-MRAM, while its current-driven write mechanism requires a relatively long write pulse and high current, increasing write latency and energy.

3.3 Task 3: Effect of Increased TMR

The MTJ resistances were changed from $R_{\text{on}} = 3\text{ k}\Omega$, $R_{\text{off}} = 6\text{ k}\Omega$ to $4\text{ k}\Omega$ and $12\text{ k}\Omega$, respectively. The main changes are shown below.

Table 10: Effect of increased MTJ resistance ratio.

Metric	Original	High TMR
Hit latency	3.763 ns	3.782 ns
Read energy	0.734 nJ	0.734 nJ
Area	0.768 mm ²	0.767 mm ²
Write latency	10.509 ns	10.517 ns
Write energy	2.063 nJ	2.064 nJ
Leakage	418.259 mW	418.259 mW

The largest change in the reported cache metrics is the small increase in hit latency, from 3.763 ns to 3.782 ns ($\approx 0.5\%$). At the array level, the increased resistance ratio changes the read path only slightly because the read current is fixed at $40\text{ }\mu\text{A}$. However, increasing the resistance contrast from $6/3 = 2$ to $12/4 = 3$ provides greater separation between the MTJ states, giving the sensing circuit a larger electrical distinction between the stored states. Thus, the device-level TMR improvement provides improved state separation, but produces only a small change in the overall cache metrics in this configuration.

3.4 Task 4: Effect of Reduced Write Current

The set/reset current was reduced from $200\text{ }\mu\text{A}$ to $100\text{ }\mu\text{A}$. The cache area remained essentially unchanged at 0.767 mm^2 , while write energy decreased significantly.

Table 11: Effect of reducing STT-MRAM write current.

Metric	200 μA	100 μA
Area	0.767 mm ²	0.767 mm ²
Write latency	10.517 ns	10.502 ns
Write energy	2.064 nJ	1.112 nJ
Leakage	418.259 mW	416.661 mW

The write energy decreases by about 46% when the write current is halved. However, the reported area does not change because the NVSim configuration keeps the cell area fixed at $54F^2$. Physically, a lower write current can allow a smaller access transistor, reducing the cell area and increasing memory density. This area benefit is therefore not captured by the present NVSim model.

https://github.com/foobar-6969/AMCAS_Assignment1